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Higher risk of spring frost under future climate change across China's apple planting regions

Renwei Chen [a], Jing Wang [a,b,*], Yang Li [c], Rui Bai [a], Mingxia Huang [d], Zhenzhen Zhang [a], Lingxuan Zhao [a], Zhenjiang Qu [e], Lu Liu [f]

[a] College of Resources and Environmental Sciences, China Agricultural University, Beijing 100193, China
[b] CMA-CAU Joint Laboratory of Agriculture Addressing Climate Change, Beijing 100193, China
[c] Chinese Academy of Meteorological Sciences, Beijing 100081, China
[d] National Climate Center, China Meteorological Administration, Beijing 100081, China
[e] Shaanxi Meteorological Training Center, Shaanxi Meteorological Service, Xi'an 710016, China
[f] Emergency and Disaster Reduction Division of Shaanxi Meteorological Service, Shaanxi Meteorological Service, Xi'an 710016, China

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ABSTRACT

Spring frost is one of major agro-meteorological disasters impacting apple production. Previous studies focused on the impacts of spring frost during the flowering period on apples, however, the frost events occurring at both pre- and post-flowering phases have significant impacts on apples. Our study employed four phenology models (parallel model, M1 model, sequential model, and growing degree hour model), along with their ensemble model driven by five global climate models (GCMs) to predict the key spring phenphases of apple across China's apple planting regions under two emission scenarios (SSP245 and SSP585) during the 2050 s (2040–2069) and 2080 s (2070–2099). A new spring frost assessment index (FI) combining frequency and intensity of spring frost was developed to evaluate the impacts of climate change on spring frost risk of apple. The results showed a large spatial variation in projected spring phenophases across China's apple planting regions. Projected budbreak (BBCH 09), initial flowering (BBCH 60), and final flowering (BBCH 69) would advance 5.6–15.9 days, 5.7–14.8 days, and 5.6–14.5 days, respectively, in regions I, II, and V of northern China apple planting regions. However, in regions III and IV of southern China apple planting regions, projected budbreak, initial flowering, and final flowering would delay –9.2–21.5 days, –7.5–20.6 days, and –6.8–20.8 days in the future climate scenarios, compared with the baseline (1980–2009). Projected last spring frost date considering the entire spring phenphases would advance 3.7–11.8 d in the future climate scenarios, compared with the baseline. Projected spring frost risk would increase in all the three growth stages including budbreak to initial flowering, flowering period, and final flowering to the last spring frost date across all the China's apple plating regions in future climate scenarios except southward China's apple plating region under the SSP585 in the 2080 s. Our study provides an important reference for early warning, prevention and mitigation of spring frost disasters, and the layout of the apple industry.

1. Introduction

Apple (*Malus domestica* L.), a widely cultivated and economically important fruit crop in temperate regions, stands as one of the four major fruit species globally (Chen et al., 2021; Wang et al., 2022). The diverse phytochemical composition of apples and their derivatives underscores their potential to impact the health of consuming populations (Hyson, 2010; Springmann et al., 2016). China is the world's largest apple producer, with the production and cultivation covering over 40 % of global output with 40–42 mil tons of apple each year mainly of the cv. 'Fuji' variety (FAO, 2023). Apple cultivation in China spans from 26°N to 47°N with various climates from the cool Northeast to warm Southwest (Ge et al., 2018).

Being a perennial crop, the growth and development of apple relies heavily on meteorological conditions and is therefore highly susceptible to climate change (Ahmadi et al., 2019; Reig et al., 2013; Yang et al., 2022; Zohner et al., 2020). Apple trees exhibit a sensitivity to low-temperature events after budbreak (Darbyshire et al., 2017; Peng

* Corresponding author at: College of Resources and Environmental Sciences, China Agricultural University, Beijing 100193, China.
E-mail address: wangj@cau.edu.cn (J. Wang).

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et al., 2018). Spring frosts could damage buds, flowers, and developing fruits (Zahradníček et al., 2024). The occurrence of spring frost often remains, but the flowering advancement now coincides so it is not the spring frost that increases, but the risk of spring frost overlapping with apple spring phenophases (Kunz and Blanke, 2022). Especially, apple flowering is highly susceptible to frost, leading to complex situations. A mild frost is beneficial as it assists in crop load regulation while severe frost is evidenced by brown discoloration at the base of the ovary to fruit loss, culminating in significant yield reduction (Aygün and Şan 2005; Dalhaus et al., 2020; Pfleiderer et al., 2019; Salazar-Gutiérrez et al., 2016; Szalay et al., 2019). Late-spring frost is recognized as the paramount agro-meteorological disaster affecting apple production globally (Liu and Sherif, 2019; Ozherelieva et al., 2023). In 2018, yield loss rate due to spring frost events reached 48 % in Shaanxi province, a staple apple production region in China. Similarly, successive spring frost events resulted in a nearly 50 % loss in apple yield in the apple growing region around Lake Geneva in New York, the United States in 2012. Moreover, a single spring frost event on April 20, 2017 led to a 24 % production loss in Europe (Wang, 2021; WAPA, 2018; Wolfe et al., 2018). Spring frost events have become a principal environmental constraint on apple production in temperate regions worldwide (Parker et al., 2021; Sun et al., 2022).

Spring frost risk of apples depends on the overlap of spring phenophases and spring frost events (Pfleiderer et al., 2019; Fujisawa and Kobayashi, 2010; Grab and Craparo, 2011). Therefore, there arises a critical necessity to precisely predict the spring phenophases of apple trees for the impact assessment of spring frost. At present, the majority of research endeavors in predicting apple spring phenophases with a single phenology model and statistical method (Darbyshire et al., 2017; Lee, et al., 2023; Luedeling et al., 2021; Ru et al., 2023a). However, recent studies showed multi-phenology model ensemble may be superior to any single model and therefore could reduce the uncertainty in assessing climate change impacts on phenology and yield of apple (Ru et al., 2023a, 2023b).

Climate change induces divergence in the variation of spring phenophases of apples in different apple planting regions (Ru et al., 2023b). On the one hand, spring frost risk may increase while the spring phenophases of apple tree occur earlier than spring frost events with the rising temperature (Chen et al., 2023; Masaki, 2020; Pfleiderer et al., 2019). On the other hand, the increase in minimum temperature may decrease the spring frost risk (Cho et al., 2021; Drepper et al., 2022a; Guo et al., 2019). Thus, rising winter and spring minimum temperatures in the context of climate change bring complex changes in spring frost risk (IPCC, 2021). Previous studies found that the response of apple trees to spring frost events under climate change were indeed region-specific (Menzel et al., 2006; Ru et al., 2023b).

Single frost risk indices, such as accumulated frost days (AFD), accumulated frost degree-days (AFDD), temperature drop rate (TDR), integrated frost stress (IFS), spring frost damage index (SFDI), have been widely used to evaluate the spring frost risk (Ru et al., 2023a, 2023b; Wang et al., 2020; Wu et al., 2022; Xiao et al., 2018). However, there is still lack of an integrated spring frost assessment index that considers both the frequency and intensity of spring frost. Moreover, the frost resistance of apple trees to freezing temperatures varies with the developmental stage, highlighting the importance of selecting the appropriate temperature thresholds in different development stages for evaluating spring frost impacts (Masaki, 2020; Salazar-Gutiérrez et al., 2016; Szalay et al., 2019). At present, a large number of studies used a constant temperature, commonly minimum temperatures of 0°C and −2°C, as temperature thresholds for evaluating frost during the flowering period (Eccel et al., 2009; Drepper et al., 2022b; Hoffmann and Rath, 2013; Guo et al., 2019; Lenz et al., 2016; Masaki, 2020; Ru et al., 2023b; Unterberger et al., 2018). However, the spring frost risk during pre- and post-flowering periods should not be neglected due to severe damage to apple production (Farajzadeh et al., 2010; Kaya et al., 2020). Therefore, it is an urgent task to fill in the knowledge gap in quantitatively assessing spring frost risk across the spring phenophases in China's apple planting regions.

The objectives of the study were to (1) assess the performance of the parallel model, M1 model, sequential model, GDH model and their ensemble in simulating spring phenophases of apple, (2) reveal the spatial and temporal variations of spring phenophases of apple based on the ensemble model, (3) evaluate spring frost risk across China's apple planting regions under future climate change scenarios based on a new spring frost assessment index. Assessments of the climate change impact on apple spring frost risk could serve as vital references for forecasting apple spring phenophases, preventing and mitigating spring frost, and optimizing the layout of the apple industry.

2. Materials and methods

2.1. Apple planting region

The delineation of China's apple planting region, with a resolution of 0.1°×0.1° (averaged for the period 2009–2011), was determined using the Spatial Production Allocation Model (SPAM2010) (Li et al., 2022; Wang et al., 2018; You et al., 2014). Grid points with apple planting areas accounting for less than 1 % of the grid area and those in provinces without records of apple production in the China Rural Statistical Yearbooks were excluded. Subsequently, 250 meteorological sites within a 50 km radius of the apple planting region regarding climate representation referring to the distribution density of meteorological sites were selected for simulating apple phenology across China's apple planting regions (Fig. 1 and S1). The regions were classified into five sub-regions: sub-region I (the Loess Plateau, including Shaaxi, Shanxi, Gansu, Ningxia, and Qinghai provinces), sub-region II (the Bohai Bay, encompassing Hebei, Liaoning, and Shandong provinces), sub-region III (the Southwest Cool Highland spanning across Xizang, Sichuan, Yunnan, and Guizhou provinces), sub-region IV (the old course of the Yellow River including Henan, Jiangsu, and Anhui provinces), and sub-region V (the specialty region covering Xinjiang, Inner Mongolia, Heilongjiang, and Jilin provinces). Table S1 showed the distribution of major apple varieties in China's apple planting region.

2.2. Historical and future climate data

Historically observed daily maximum and minimum air temperatures (°C), rainfall (mm), and sunshine hours (h) from 1979 to 2017 from China Meteorological Administration (CMA). Daily solar radiation ($MJ{\cdot}m^{-2}{\cdot}d^{-1}$) was calculated from sunshine hours using the Angstrom equation (Wang et al., 2015b). Five GCMs from the Coupled Model Inter-comparison Project Phase 6 (CMIP6) that exhibited good performance in replicating China's historical frost days and spring frost dates in China (with R^2 more than 0.8 and RMSE less than 0.72 days) were used under two future Shared Socio-Economic pathways including a moderate emission scenario (SSP245) and a strong emission scenario (SSP585) (Table S2, Zhong and Qian, 2022). Projected daily maximum and minimum temperatures (°C), rainfall (mm), and solar radiation ($MJ{\cdot}m^{-2}{\cdot}d^{-1}$) for each study site from 1981 to 2100 were statistically downscaled with NWAI–WG model from monthly gridded climate data simulated by the five GCMs under the two emission scenarios (Text S1) (Liu and Zuo, 2012). The method was used widely in generating daily climate data for climate change impact assessment due to high computational efficiency (Huang et al., 2020, 2021, 2022).

Fig. S1 showed the spatial distribution of annual average temperature, annual precipitation, and annual solar radiation averaged from the historical period (1980–2009) across China's apple planting regions. Temperature and precipitation increased from north to south while solar radiation decreased from north to south. Regions III and IV had higher temperature than the northern region. Fig. 2 showed the changes in apple growing season (April to October) temperature, precipitation, and solar radiation in the future climate scenarios averaged from the entire

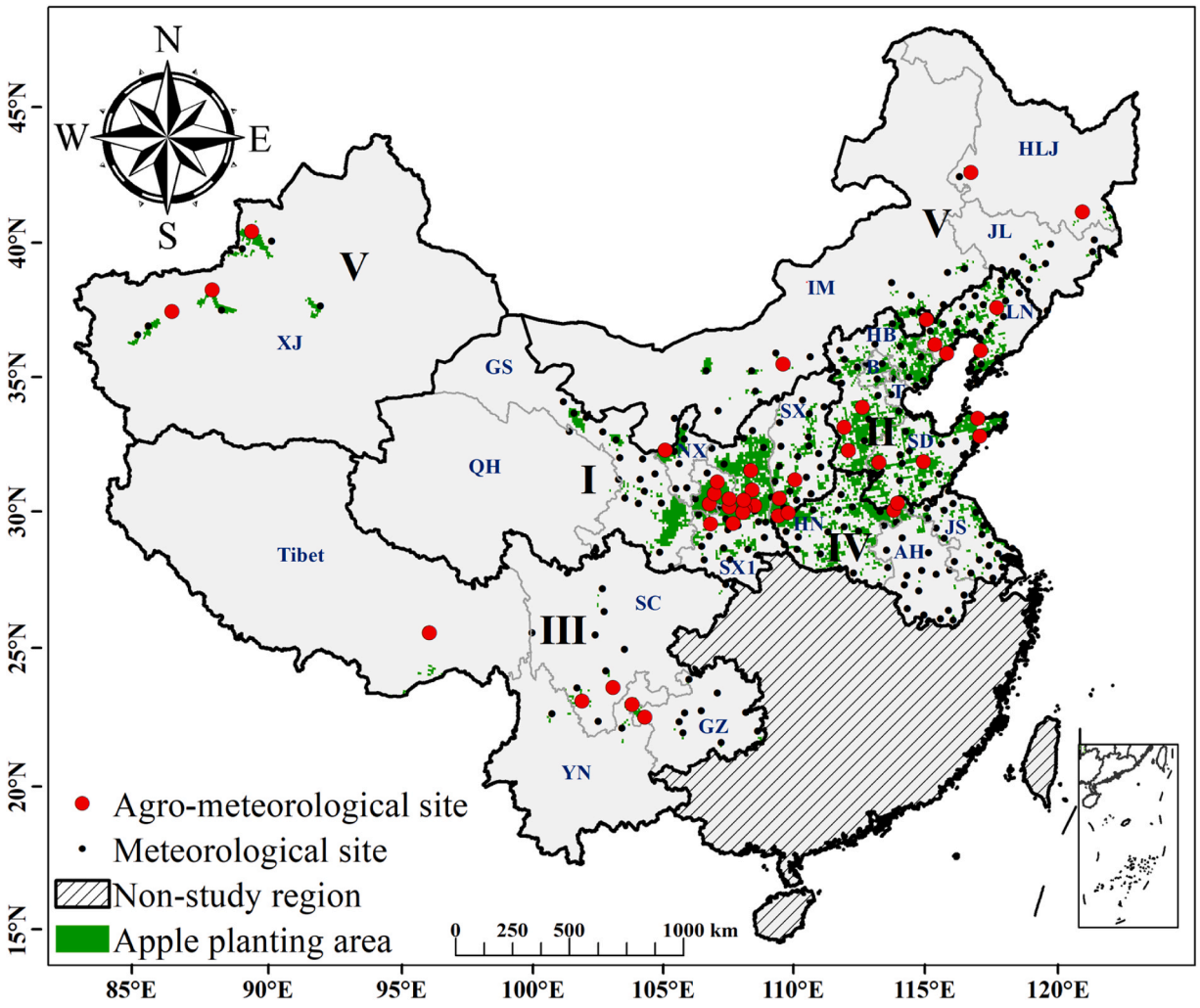


Fig. 1. The apple planting sub-regions in China (Sub-region I: the Loess Plateau, Sub-region II: the Bohai Bay, Sub-region III: the Southwest Cool Highland, Sub-region IV: the old course of the Yellow River, Sub-region V: the specialty region) and the distribution of meteorological and experimental sites containing data for the calibration and validation of apple phenology models. The abbreviations of the provinces mean: Xinjiang (XJ), Qinghai (QH), Gansu (GS), Ningxia (NX), Shaanxi (SX1), Shanxi (SX), Hebei (HB), Beijing (B), Tianjin (T), Heinan (HN), Shandong (SD), Liaoning (LN), Jilin (JL), Heilongjiang (HLJ), Inner Mongolia (IM), Jiangsu (JS), Anhui (AH), Sichuan (SC), Guizhou (GZ), and Yunnan (YN).

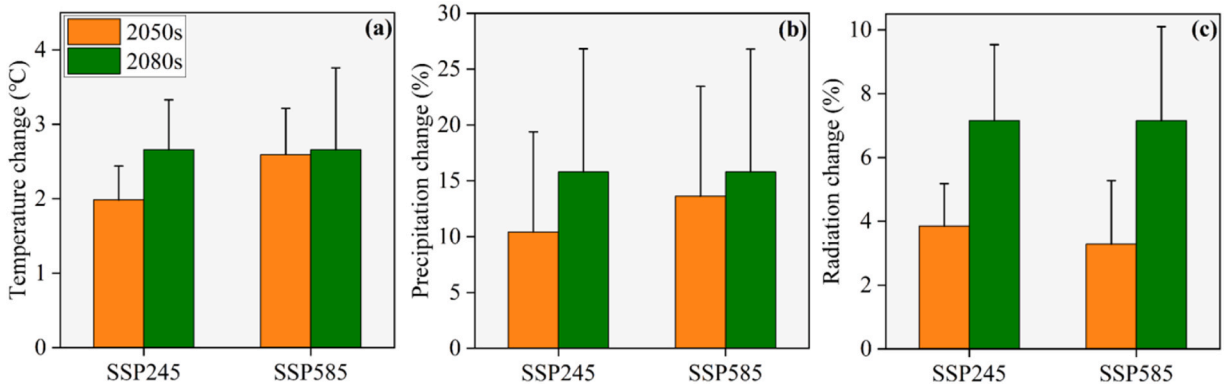


Fig. 2. Change in apple growing season (April to October) average temperature (a), total precipitation (b), and total solar radiation (c), in the 2050 s (2040–2069) and 2080 s (2070–2099) compared with the baseline (1980–2009) under the SSP245 and SSP585 averaged from entire China's apple planting region. Error bars show standard deviation across five GCMs.

China's apple planting region. Apple growing season temperature, precipitation, and solar radiation were projected to increase in the future climate scenarios, compared with the baseline (1980–2009), which was commonly selected to represent historical climate (Cammarano et al., 2022; Huang et al., 2020, 2021, 2022). The climate data was bias-adjusted referencing Text S1 and projected climate change were

consisted with previous studies (Lu et al., 2022; Wu et al., 2019).

2.3. Apple phenology model

Parallel model, M1 model, sequential model, and growing degree hour (GDH) model were used to simulate the budbreak, initial flowering, and final flowering of representative apple varieties. These models performed well in capturing the impacts of climate change on spring phenology of apple over the world (Blümel and Chmielewski, 2012; Wang et al., 2021). In one-phase models, the GDH model focused solely on the forcing stage of fruit dormancy, utilizing an asymmetric curve for precise calculation of daily thermal time. In contrast, the M1 model incorporated a thermal time approach alongside daylength correction. For two-phase models, parallel model and sequential model considered both chilling requirement and thermal time. Their distinction was that the ecological dormancy and endodormancy processes occurred simultaneously in the parallel model while the endodormancy process completes first followed by the ecodormancy process in the sequential model. Detailed information regarding the phenology models and parameters of major apple varieties were shown in Text S2 and Table S3, respectively.

Observed phenological data (e.g., budbreNorwaak (BBCH 09), initial flowering referring to the first flower opening (BBCH 60), and final flowering dates signifying all petals fallen (BBCH 69)) of major apple varieties were obtained from agro-meteorological sites and used to calibrate the phenology models (Tables S4). The parameters of major apple varieties in the phenology models were determined by the trial-and-error method.

2.4. Spring frost risk assessment index

A new spring frost risk assessment index (FI) was conducted by combining frost frequency (F_f) and frost intensity (AFDD):

$$\mathrm{FI_i}(^{\circ}\mathrm{C{\cdot}d}) = F_{\mathrm{fi}} \times \mathrm{AFDD}_i \quad (1)$$

$$\mathrm{FI_{fi}}(\%) = \frac{\mathrm{AFD}_i}{L_{\mathrm{i}}} \times 100 \quad (2)$$

where i represents three growth stages, from budbreak to initial flowering (phase_1), flowering period (phase_2), and final flowering to the last spring frost date after final flowering (phase_3). L_{i} represents the length of each growth stage.

AFDD (accumulated frost degree-days with minimum temperature less than critical temperature threshold) and AFD (accumulated frost days with minimum temperature less than critical temperature threshold) were calculated as follows (Xiao et al., 2018):

$$\mathrm{AFDD}_i = \sum_{\mathrm{phenophase}_i}^{\mathrm{phenophase}_{i+1}} \mathrm{Max}(T_h - T_{\min}\ ,\ 0) \quad i = 1,\quad 2, and 3 \quad (3)$$

$$\mathrm{AFD}_i = \sum_{\mathrm{phenophase}_i}^{\mathrm{phenophase}_{i+1}} \mathrm{Frostdays} \quad (T_{min} < T_h) \quad i = 1,\quad 2, and 3 \quad (4)$$

where, phenophases represent budbreak ($\mathrm{phenophase}_1$), initial flowering ($\mathrm{phenophase}_2$), final flowering ($\mathrm{phenophase}_3$), and the last spring frost date after final flowering ($\mathrm{phenophase}_4$), respectively. $T_{\min}$ is the minimum temperature. T_h is the temperature threshold of frost for each growth stage. –2, –1, and –0.5°C were employed to represent T_h for phases 1, 2, and 3, respectively, according to grade of crop frost damage (China Meteorological Administration, 2008) issued by Chinese Meteorological Industry Standards. The last spring frost event during each phases was calculated based on the temperature threshold of frost for each phase. The latest date for the last spring frost event was determined as the last spring frost date. The risk distribution of spring frost for apple during the historical period was divided using natural breaks classification into five classes: high, moderately low, middle, moderately high, and high. The delineation of critical values for historical period frost risk was applied to future climate change scenarios.

2.5. Long-term simulation setting

Long-term simulations of apple phenology were conducted for the baseline period (1980–2009), the 2050 s (2040–2069), and the 2080 s (2070–2099). To compare the simulations in the baseline, the 2050 s, and the 2080 s, we used all climate projection data downscaled by five GCMs from CMIP6. The projected day of year (DOY) for spring phenophases during the historical and future periods was based on an ensemble model derived from the average of four phenology models. The apple varieties were held unchanged during the historical and future periods.

2.6. Statistical metrics for phenology model evaluation

The adjusted coefficient of determination (R^2), root mean square error (RMSE), and normalized root mean square error (NRMSE) between observed and simulated data were used to evaluate the performance of phenology models in simulating phenophase (budbreak, initial flowering, and final flowering):

$$R^2 = \frac{\left[\sum_{i=1}^{n}(X_i - \overline{X})(Y_i - \overline{Y})\right]^2}{\sum_{i=1}^{n}(X_i - \overline{X})^2 \sum_{i=1}^{n}(Y_i - \overline{Y})^2} \quad (5)$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(Y_i - X_i)^2}, \quad (6)$$

$$NRMSE(\%) = \frac{\sqrt{\frac{1}{n}\sum_{i=1}^{n}(Y_i - X_i)^2}}{\overline{X}} \times 100, \quad (7)$$

where X_{i} and Y_{i} represent the observed and simulated date of budbreak, initial flowering, and final flowering, respectively, $\overline{X}$ and $\overline{Y}$ represent the mean of observed and simulated date of budbreak, initial flowering, and final flowering, respectively, n is the number of observation.

2.7. Spatial analysis

The spring phenophase of apple, frequency (AFD), intensity (AFDD), and risk of spring frost were spatially interpolated by applying the inverse distance weighting (IDW) interpolation method, which is a popular interpolation method for large areas and moderate station density (He et al., 2017).

3. Result

3.1. Model performance evaluation

Fig. 3 showed the comparisons of simulated and observed budbreak, initial flowering, and final flowering for major apple varieties at the agro-meteorological sites of apple planting regions. In general, parallel model, sequential model, M1 model, and GDH exhibited good performance in simulating spring phenophases of apple. RMSE values between observed and simulated spring phenophases were below 6.2 days for both calibration and validation periods. Furthermore, the ensemble model further improved the simulation accuracy. Observed and simulated spring phenophases showed good performance with RMSEs between observed and simulated budbreak, initial flowering, and final flowering were 5.6, 4.6, and 5.2 days, respectively for calibration periods, and 5.9, 4.2, and 4.7 days, respectively for the validation period. R^2 between observed and simulated budbreak, initial flowering, and final flowering for calibration period was 0.85, 0.84, and 0.79, respectively, while it for validation period was 0.81, 0.87, and 0.83,

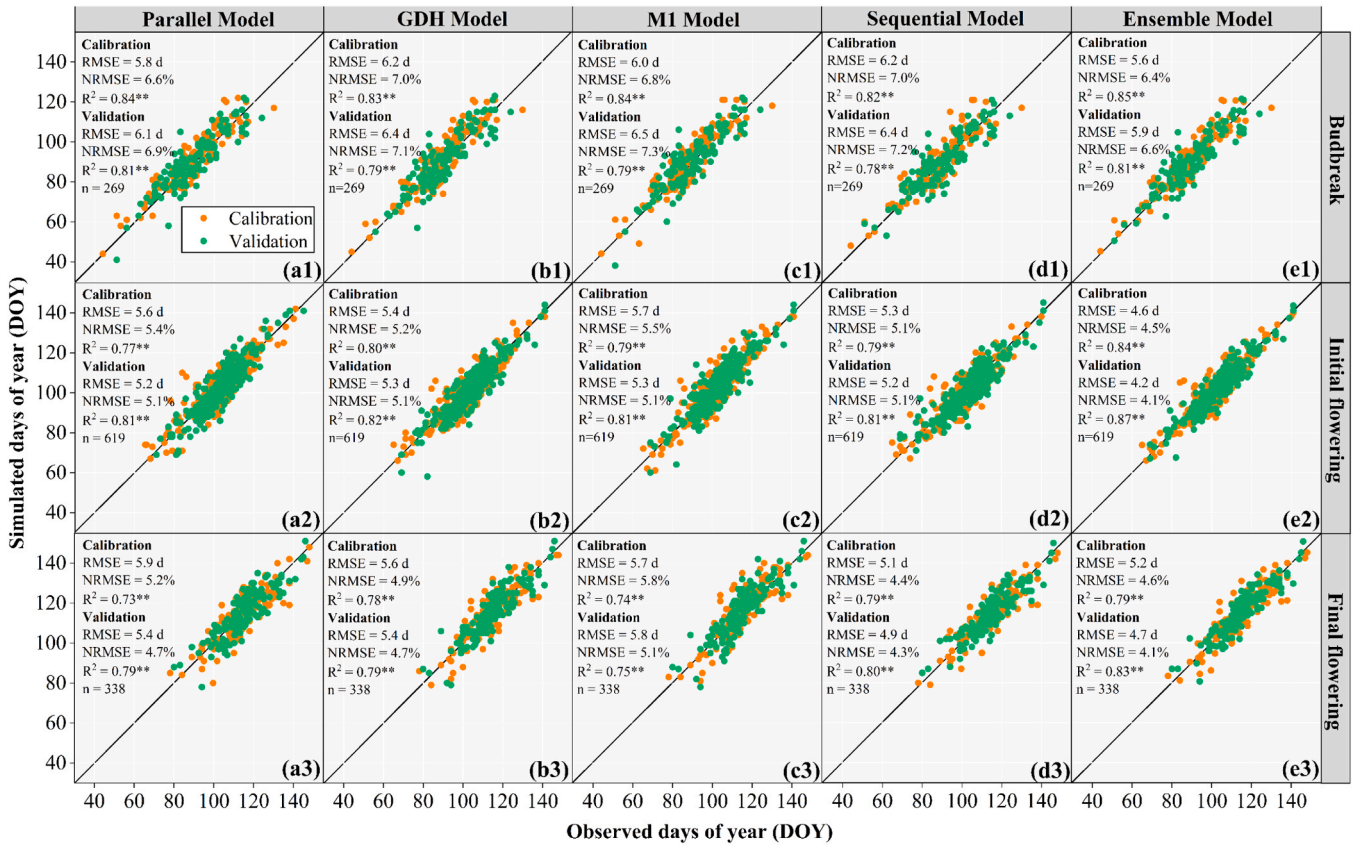


Fig. 3. Performance of individual phenology models (parallel model: a1-a3, GDH model: b1-b3, M1 model: c1-c2, sequential model: d1-d3) and ensemble model (e1-e3). ** represents $P < 0.01$.

respectively.

3.2. Change in spring phenophases of apple

Fig. 4 showed the spring phenophases simulated by the ensemble model. Simulated dates (DOY) of budbreak, initial flowering, final flowering, and last spring frost were 95.0 ± 10.2, 110.8 ± 10.4, 119.9 ± 9.8, and 113.2 ± 12.0 in the historical period (1980–2009), respectively. Simulated spring phenophases and last spring frost dates followed a quasi-latitudinal distribution, i.e. earlier spring phenophases and last spring frost dates with warmer climate. Averaged last spring frost dates normally occurred after initial flowering, except regions II and IV where averaged last spring frost date occurred before the budbreak. Largest difference in simulated spring phenophases of apple between sites occurred in regions III and V while the largest difference in simulated last spring frost date of apple between sites occurred in regions I and V across the China's apple planting regions. Projected values of spring phenophases and the last spring frost date and their changes in the historical period were shown in Table S5.

In general, spring phenophases of apple were projected to advance in the dominant apple planting regions of northern China but they were projected to delay in the southern regions, compared with baseline. Under higher emission scenarios (SSP585), spring phenophases were projected to advance more in regions I, II, and V compared to lower emission scenarios (SSP245), whereas the opposite trend was projected to occur in regions III and IV. Particularly in the 2080 s under the SSP585 scenario, in warmer climate regions in China, including the southern areas of region I, as well as most parts of regions III and IV, exhibited more pronounced delays in spring phenology compared to other scenarios.

Compared with the baseline, projected budbreak, initial flowering, and final flowering would advance 5.6–10.3 days, 5.7–9.5 days, and 5.6–9.1 days respectively under SSP245, and 8.3–15.9 days, 8.3–13.9 days, and 8.1–14.5 days respectively under SSP585 in regions I, II, and V. However, projected budbreak, initial flowering, and final flowering would delay –8.9–2.0 days, –6.9–1.3 days, and –6.4–1.3 days respectively under SSP245, and –9.2–21.5 days, –7.5–20.5 days, and –6.8–20.8 days respectively under SSP585, in regions III and IV compared with the baseline. Projected last spring frost date would advance 4.6–7.4 days and 6.6–10.4 days, respectively, under SSP245 and SSP585, compared with the baseline. The spring phenology was projected to advance more compared with the last spring frost date under the future climate scenario, which suggested the higher spring frost risk in the future. Largest variation in projected spring phenophases between GCMs were projected to occur in region III across the China's apple planting regions. Table S6 showed projected values of spring phenophases and the last spring frost dates and their changes in the future climate scenarios.(Fig. 5)

3.3. Change in apple spring frost risk

Fig. 6 showed that high spring frost risk regions were prevalent in northern region I, northwestern region V, and southeastern region III while regions II and IV predominantly experienced low spring frost risk. The stage from budbreak to initial flowering ($phase_1$) exhibited the highest spring frost risk, followed by the flowering period ($phase_2$), and the stage from final flowering to the last spring frost date after final flowering ($phase_3$). Simulated average spring frost risk index of $phase_1$, $phase_2$, and $phase_3$ was 0.05 ± 0.07 °C·d, 0.008 ± 0.013 °C·d, and 0.005 ± 0.014 °C·d, respectively in the historical period. Projected values of spring frost risk classification and distribution in the historical period were shown in Tables S7-8. All regions were projected to exhibit high

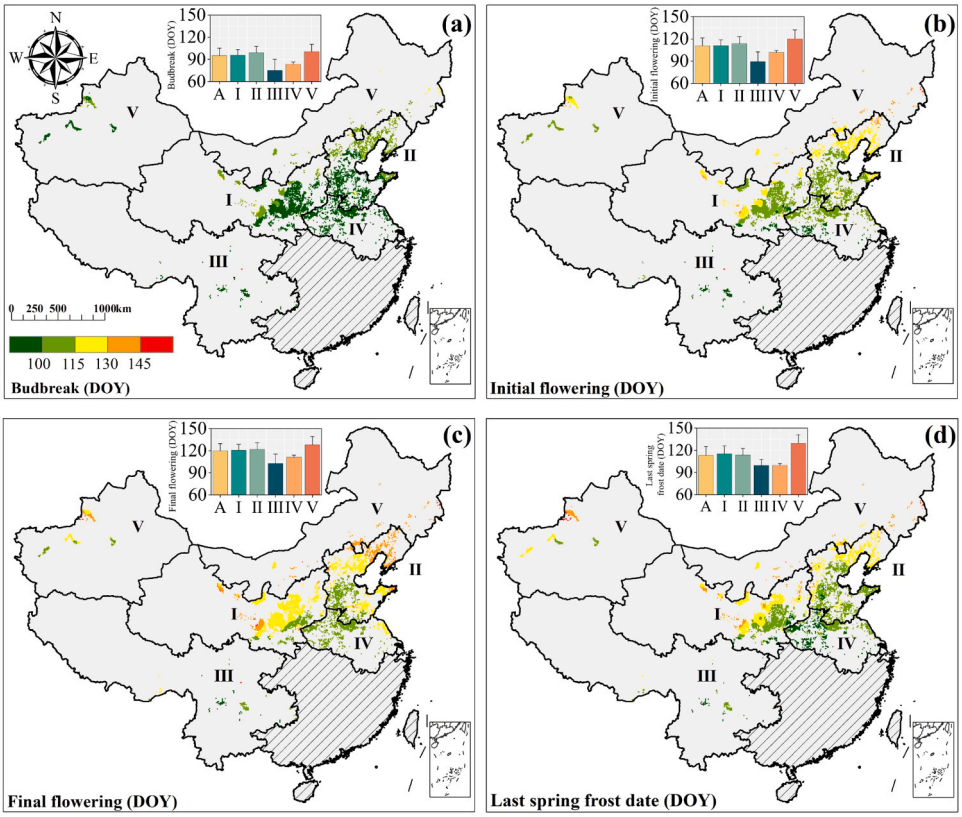


Fig. 4. Simulated apple spring phenphases (budbreak (a), initial flowering (b), and final flowering (c)) and last spring frost date (d) in the historical period. Panels inserted in Fig. 4a-d show the dates of budbreak, initial flowering, final flowering, and last spring frost date averaged from all the regions (A) and in each region. Error bars show standard deviation across each sub-region.

variability in spring frost risk due to large differences between sites. AFD and AFDD have similar spatial distribution with spring frost risk index across China's apple planting regions (Fig. S2 and Table S9).

Fig. 7 indicated that high spring frost risk regions were prevalent in northern regions I and V and western region II while most parts of regions III and IV were low spring frost risk regions. Higher spring frost risk was projected to occur in northwestern parts of China's apple planting region and western part of region II in future climate scenarios. Similar to historical climate period, $phase_1$ was projected to exhibit the highest spring frost risk followed by $phase_2$ and $phase_3$. Projected spring frost risk would increase in regions I, II, and V while decrease in most southern regions in future climate scenarios, compared with baseline. Under higher emission scenarios (SSP585), spring frost risk index would increase more in regions I, II, and V compared to lower emission scenarios (SSP245). Largest variation in projected spring frost risk between GCMs was projected to occur in regions I and V across China's apple planting regions. Projected values of spring frost risk classification and distribution in the future climate scenarios was shown in Table S10.

4. Discussions

4.1. Simulation accuracy of apple phenology models

In comparison to annual crops, the phenology of perennial fruit typically necessitates the consideration of chilling requirements (Drepper et al., 2022a; Eccel et al., 2009; Rodríguez et al., 2021). Insufficient chilling requirements resulted in delayed foliation, diminished flowering quality and uniformity, ultimately leading to reduced production (Duchêne and Schneider 2005; Funes et al., 2016). Therefore, assessing the impacts of climate change on apple phenology needs to consider the change in chilling requirements (Laube et al., 2014; Leolini et al., 2018; Morin et al., 2009; Pope et al., 2013). At present, one-phase and two-phase phenology models were frequently employed for simulating phenology of temperate fruit trees (Guo et al., 2019; Wang et al., 2022). One-phase models only considered ecodormancy while two-phase models included both ecodormancy and endodormancy to simulate apple development process (Lang et al., 1987). Due to different mechanisms among the models, there was a large uncertainty in simulated apple spring phenophases with a single model (Wang et al., 2015a). Therefore, our study included multiple phenology models considering thermal time, chilling requirement, and photoperiod. Previous study found that one-phase phenology models performed well in the region with fulfilled chilling requirements (e.g. Norway) (Meld, 2013). In the one-phase phenology models, the GDH model emerged as one of the most proficient forcing models for numerous temperate fruit trees, while M1 model considered photoperiod to improve simulation accuracy for various temperate fruit trees (Blümel and Chmielewski, 2012; Fraga et al., 2019; Gu, 2016; Kaufmann and Blanke, 2019). In the two-phase phenology models, sequential model performed well in simulating apple flowering worldwide while parallel model considered chilling requirements performed well in simulating apple initial- and final flowering dates (Darbyshire et al., 2017; Wang et al., 2021). The simulation results proved that the ensemble model was superior to any individual model, especially in regions with significant variations in meteorological conditions (Zheng and Zhang, 2023). Therefore, ensemble model could be encouraged to extensive application in climate change impact assessment to reduce the uncertainties of simulated phenology (Ru et al., 2023a, 2023b; Zheng and Zhang, 2023).

4.2. Impacts of climate change on spring phenology of apple

In this study, we assessed the apple spring phenology using the SPAM

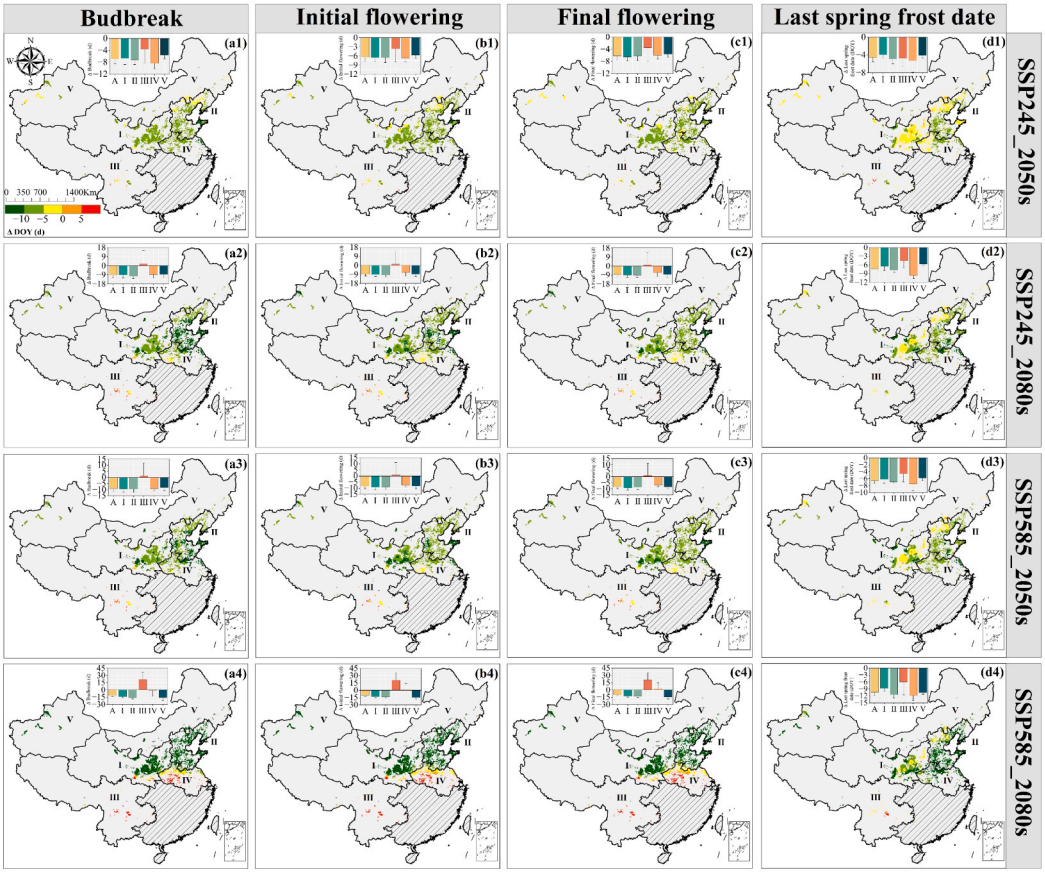


Fig. 5. Projected changes of spring phenology based on the ensemble model in China's apple planting regions, in the 2050 s (2040–2069) and 2080 s (2070–2099), compared with the baseline (1980–2009). Projected changes in budbreak, initial flowering, final flowering, and last spring frost date in the 2050 s (a1, b1, c1, and d1) and 2080 s (a2, b2, c2, and d2) under the SSP245 scenario. Projected budbreak, initial flowering, final flowering, and last spring frost date in the 2050 s (a3, b3, c3, and d3) and 2080 s (a4, b4, c4, and d4) under the SSP585 scenario. Panels inserted in Fig. 1 show the dates of budbreak (a1-a4), initial flowering (b1-b4), final flowering (c1-c4), and last spring frost date (d1-d4) averaged from all the regions (A) and in each subregion. Error bars show standard deviation across five GCMs.

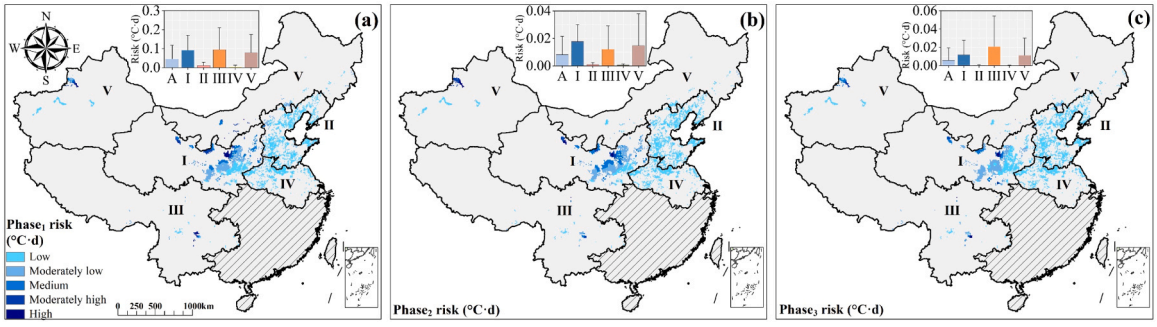


Fig. 6. Spring frost risk index of budbreak to initial flowering (phase$_1$), flowering period (phase$_2$), and final flowering to the last spring frost date after final flowering (phase$_3$) in the historical period. Panels inserted in the figure show the risk of phase$_1$, phase$_2$, and phase$_3$ (a-c) averaged from all the regions (A) and in each region. Error bars show standard deviation across each sub-region.

grid data of China's apple planting region by considering small apple planting regions from northeastern China. There was a large spatial heterogeneity in the projected spring phenology of apples by the ensemble model across China's apple planting regions. Compared with the previous simulation study of apple phenology in China, there was similar spatial change in simulated spring phenology (Ru et al., 2023a, 2023b; Wang et al., 2021). Projected spring phenophases advanced in the temperate apple planting regions where the chilling demand could be met (Drkenda et al., 2018; Unterberger et al., 2018; Vitasse et al., 2018; Yang et al., 2021). In the warmer climate regions across China's apple planting regions, such as region III and southern region IV, spring phenology would be delayed due to insufficient chilling requirements caused by rising winter and spring temperatures (Guo et al., 2019). In this study, different warming degrees between high and low latitudes would result in a divergence in the change of spring phenology in future climate change scenarios. The advancement in spring phenophases were projected to occur in regions I, II, and V. However, decreased chilling requirements due to climate warming would even result in a delay in the spring phenophases, particularly in the southern region I, most parts of regions III and IV albeit with high uncertainty between GCM projections (Fu et al., 2015; Funes et al., 2016; Guo et al., 2019; Harrington et al., 2010; Jeong et al., 2011). Delayed spring phenology due to insufficient

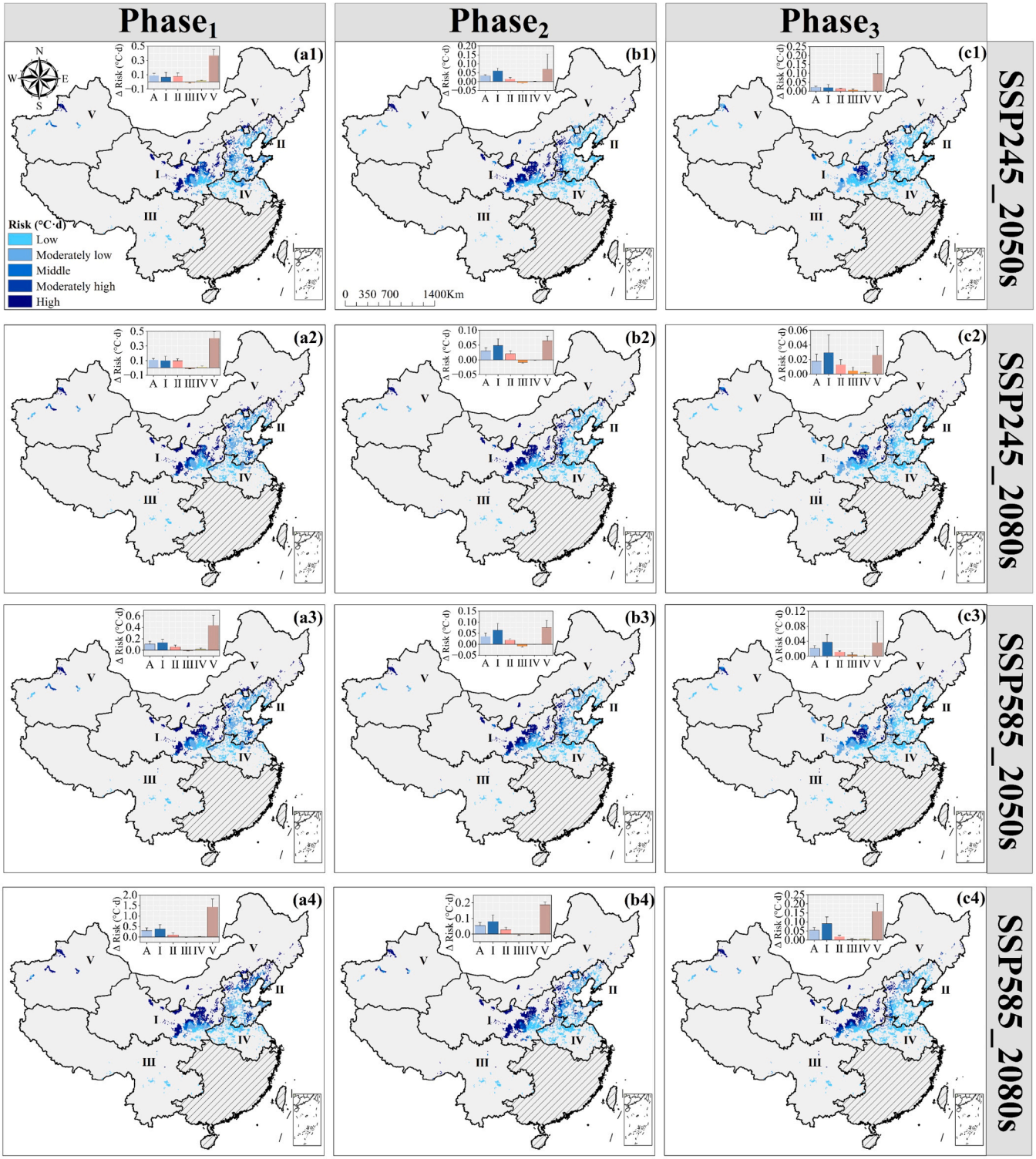


Fig. 7. Projected spring frost risk based on frost risk index in China's apple planting regions, in the 2050 s (2040–2069) and 2080 s (2070–2099) by natural breaks classification. Projectebd risk index in phase$_1$, phase$_2$, and phase$_3$ in the 2050 s (a1, b1, and c1) and 2080 s (a2, b2, and c2) under the SSP245 scenario. Projected change of phase$_1$, phase$_2$, and phase$_3$ in the 2050 s (a3, b3, and c3) and 2080 s (a4, b4, and c4) under the SSP585 scenario. Panels inserted in the figure show the risk index of phase$_1$ (a1-a4), phase$_2$ (b1-b4), and phase$_3$ (c1-c4) averaged from all the regions (A) and in each subregion. Error bars show standard deviation across five GCMs.

chilling requirements in the future was also projected in the temperate apple planting regions such as Spain, California, and Portugal (Rodríguez et al., 2021; Baldocchi and Wong, 2008; Delgado et al., 2021; Fraga and Santos, 2021). Moreover, changes in spring phenophases are more pronounced in China compared with other temperate apple planting regions (Darbyshire et al., 2013; Grab and Craparo, 2011; Karami and Asadi, 2017; Legave et al., 2013; Masaki, 2020; Unterberger et al., 2018; Vitasse et al., 2018), which possibly attributed to the more significant warming observed in mid- to high-latitude of China (Ji et al., 2014). Moreover, our study quantified the change in the last spring frost

date based the temperature thresholds for different spring growth stages to improve the assessment precision of apple frost risk (Unterberger et al., 2018).

4.3. Change in apple spring frost risk in the future

The spring frost risk depends on the dynamic change of the fruit tree's phenophases and the timing of spring frost events (Guo et al., 2019; Masaki, 2020). In regions I and II, northern region IV, and region V, apple flowering was projected to occur earlier than spring frost events in the future (Lamichhane, 2021; Unterberger et al., 2018; Vitasse et al., 2018; Yang et al., 2021). In some warm regions, flowering was projected to be delayed due to inadequate chilling requirements leading to the decline in frost risk (Cho et al., 2021; Drepper et al., 2022b; Guo et al., 2019). In this study, the frequency and intensity of spring frost were greater in the high emission scenario (SSP585) compared to the low emission scenario (SSP245), especially in 2080 s. This was because spring phenophases of apple would advance more under SSP585 than under SSP245. Projected spring frost risk would increase in regions I, II, and V but decrease in regions III and IV. However, the climate change impacts resulting in the change of spring frost were region-specific. Here, our study found that spring frost frequency would decrease while the spring frost intensity would increase in the parts of southern region I, northeastern region II, and region V, which was consistent with the previous study (Ru et al., 2023b). Moreover, the cold resistance of apples decreased with dehardening or acclimation. Utilizing the same indicator across different spring growth stages would postpone the last spring frost date and therefore lead to an overestimation of spring frost risk (Masaki, 2020). However, although the previous study projected the similar risk degree between different climate change scenarios (Ru et al., 2023b), our study projected the increasing frequency and intensity of spring frost of apples with periods (Figs. S3–4 and Tables S11-12). The difference was due to the inconsistency in the choice of phenology models, GCMs, the threshold for spring frost, and apple planting regions. Additionally, Northeastern Yunnan province and western Guizhou province was highly frost-prone areas in the historical period. Our results were consistent with the actual observational records according to the survey of China Meteorological Administration Apple Centre. Although AFD and AFDD would decrease with the time, the most severe frost damage was reported during the flowering period (Chen et al., 2023; Zhu et al., 2021), which might be attributed to organ vulnerability to frost (Charrier et al., 2013, 2015; Guillaume et al., 2018).

Despite previous studies have explored the frequency and intensity of spring frost, a single indicator was insufficient to fully describe the extent of spring frost occurrence (Ru et al., 2023a, 2023b; Xiao et al., 2018). Therefore, the study conducted a new spring frost risk assessment index by considering frost frequency and intensity simultaneously. The degree of frost risk was projected to decrease with apple spring growth stages. In the future, the spring frost-sensitive regions would increase in northern China while almost no frost risk would occur in southward China.

4.4. Study limitations

There were several limitations in our study. Firstly, the latest available China apple planting region for simulation was averaged from 2009 to 2011 based on a temperate fruit planting region dominated by apples. Updated data for China's apple planting region is needed urgently to assess the climate change impacts on apple phenology more accurately. Secondly, our study did not consider the difference of the temperature thresholds for spring frost between apple varieties because Fuji-based apple varieties account for 70 % of total apple production in China. However, the temperature thresholds for spring frost may vary across different apple varieties, tree ages, and management options (Drepper et al., 2022a; Szalay et al., 2019). The results may overestimate the spring frost risk in the regions where cold-resistant apples were planted. However, there is lack of index reflecting the cold-resistance of different apple varieties, which needs to be further studied. Thirdly, more observation data on spring phenophases of apple varieties should be collected to further validate the ensemble model, particularly in southward China's apple planting regions.

5. Conclusion

Althogh sequential model, parallel model, M1 model, and GDH perform well in evaluating apple spring phenophases, their ensemble model could further enhance the simulation accuracy of apple phenology. Overall, in future climate change scenarios, the spring phenophases would advance in northern China while delay in southern China. Spring frost risk of apple would be region-specific dependent upon the timing of spring phenophases and spring frost events across China's apple planting regions. The frequency, intensity, and risk of spring frost of apples would increase in all the spring growth stages under the future climate change scenarios in northern China's apple planting regions, especially in northwestern China while the result would be the opposite in southern China's apple planting regions. Intensified spring frost risk calls for effective adaptation strategies to alleviate the frost damage in staple China's apple planting regions.

CRediT authorship contribution statement

Renwei Chen: Writing – review & editing, Writing – original draft, Formal analysis, Data curation. **Jing Wang:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Zhenjiang Qu:** Resources, Funding acquisition, Conceptualization. **Lu Liu:** Resources, Funding acquisition, Conceptualization. **Zhenzhen Zhang:** Methodology, Conceptualization. **Lingxuan Zhao:** Methodology, Conceptualization. **Rui Bai:** Methodology, Conceptualization. **Mingxia Huang:** Methodology, Conceptualization. **Yang Li:** Methodology, Conceptualization.

Declaration of Competing interest

The authors declare no competing interests.

Data Availability

Data will be made available on request.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.eja.2024.127288.

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